Throughput QOS

2026-03-12-11-18-56

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

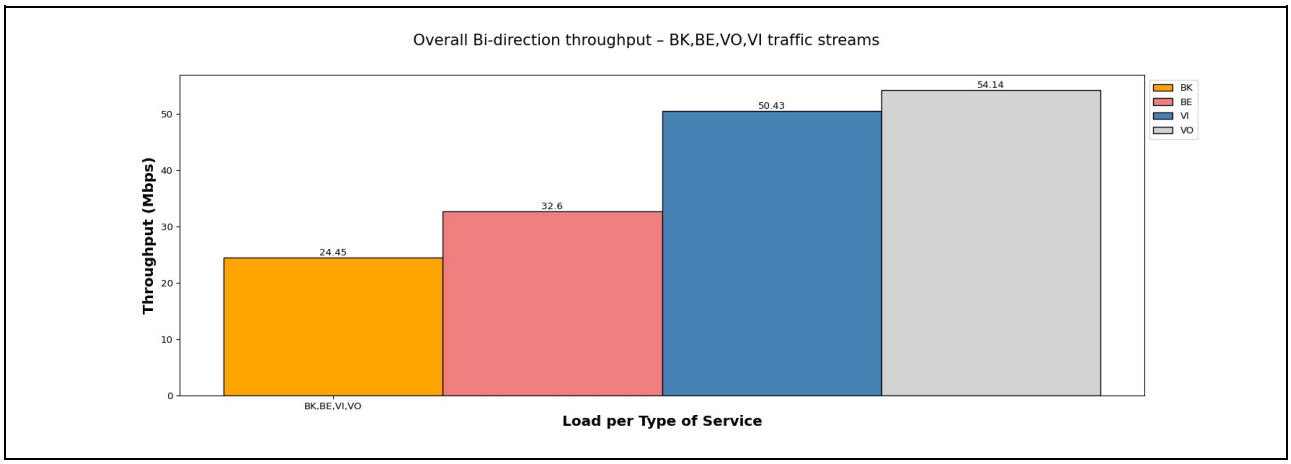
Test Configuration	Number of Stations	8
	AP Model	Cisco
	SSID_2.4GHz	NETGEAR_2G_wpa2
	SSID_5GHz	NETGEAR_5G_wpa2
	SSID_6GHz	None
	Traffic Duration in hours	0.02
	Security_2.4GHz	wpa2
	Security_5GHz	wpa2
	Security_6GHz	None
	Protocol	TCP
	Traffic Direction	Bi-direction
	TOS	['BK', 'BE', 'VI', 'VO']
	Per TOS Load in Mbps	Upload:100.0 Mbps,Download:100.0 Mbps

Overall Bi-direction throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 100.0 Mbps-upload	Throughput for Load 100.0 Mbps-download
4 + 4	BK : 11.28, BE : 15.14, VI: 26.88, VO: 29.89	BK : 13.17, BE : 17.46, VI: 23.55, VO: 24.25

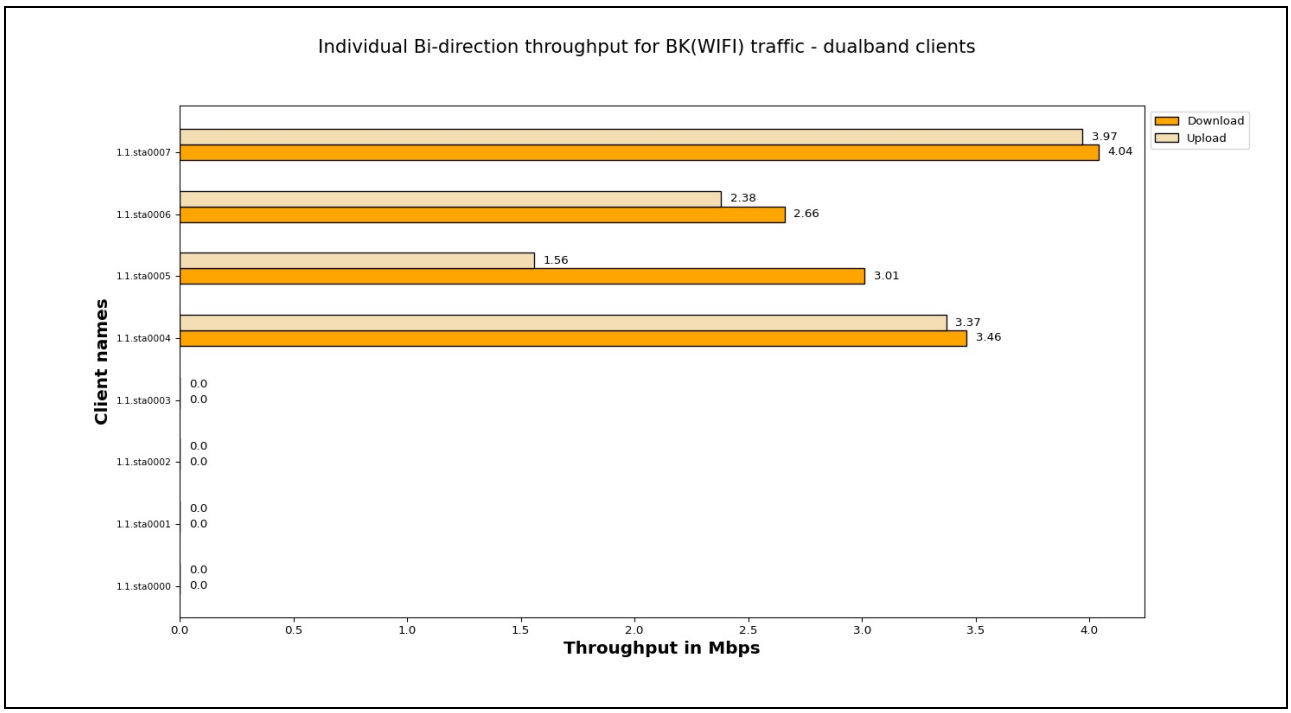
Overall Bi-direction throughput for 8 clients for 100.0 Mbps-download band with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loads per station – 100.0 Mbps-download



Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 8 clients running BK (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

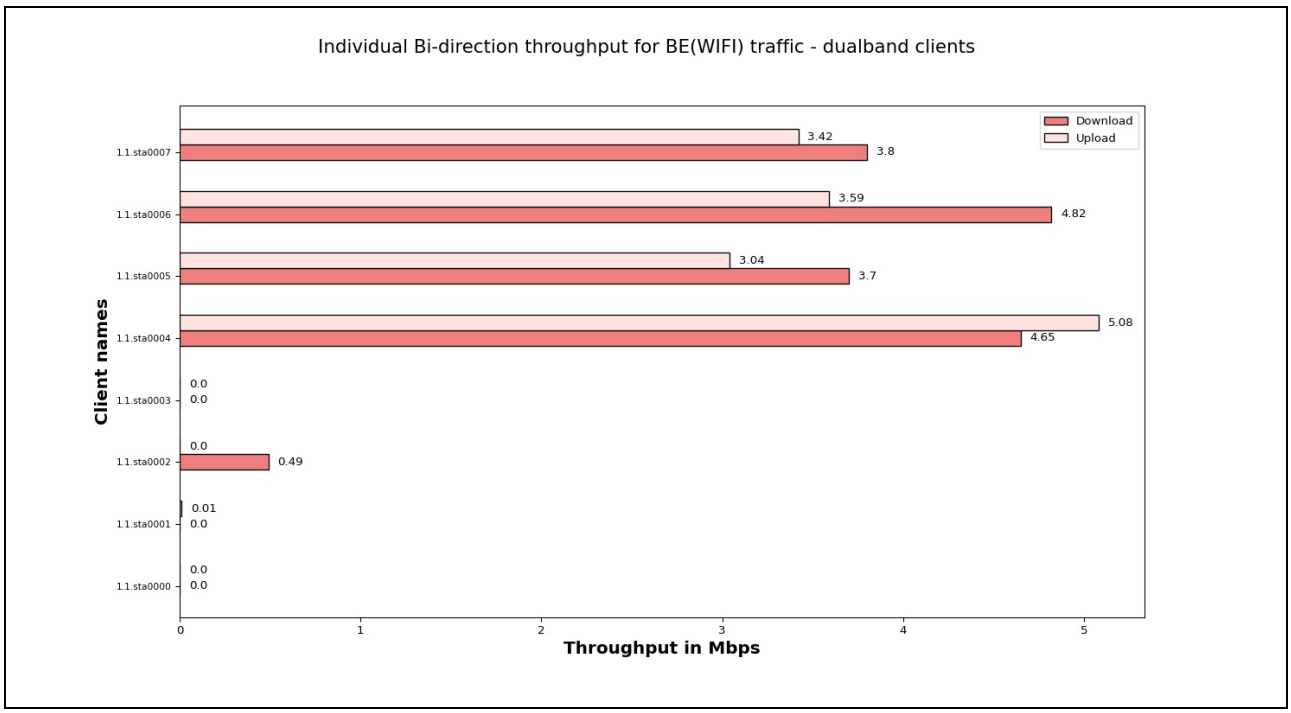


TOS : Background

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:6f:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	100.000000	100.000000
1.1.sta0001	04:f0:21:94:b7:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	83.333333	83.333333
1.1.sta0002	04:f0:21:94:93:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	100.000000	87.500000
1.1.sta0003	04:f0:21:94:68:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	75.000000	75.000000
1.1.sta0004	00:0a:52:07:03:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	3.37	3.46	15.537849	8.637874
1.1.sta0005	00:0a:52:07:be:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	1.56	3.01	26.062323	7.473310
1.1.sta0006	00:0a:52:07:71:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	2.38	2.66	19.937695	7.829181
1.1.sta0007	00:0a:52:07:a5:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	3.97	4.04	12.500000	7.942238

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 8 clients running BE (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

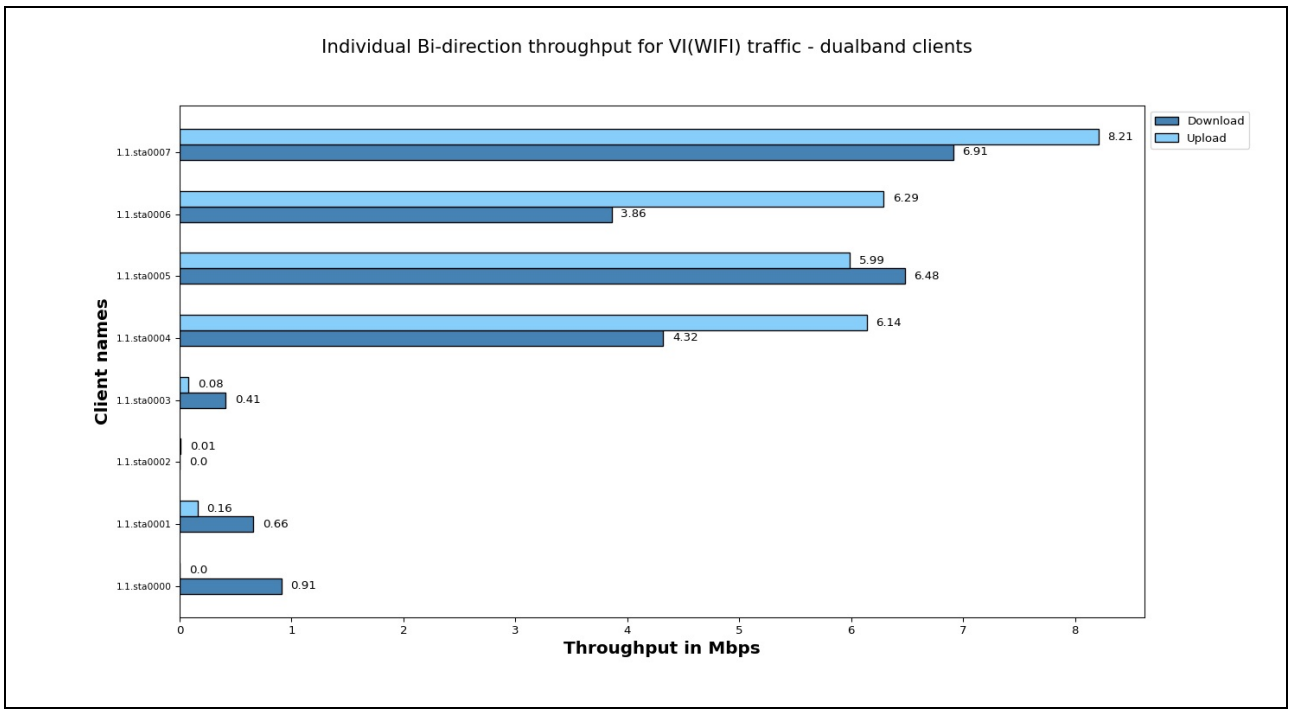


TOS : Besteffort

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:6f:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	100.000000	62.500000
1.1.sta0001	04:f0:21:94:b7:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.01	0.00	87.500000	43.750000
1.1.sta0002	04:f0:21:94:93:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.49	33.333333	24.000000
1.1.sta0003	04:f0:21:94:68:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	100.000000	80.000000
1.1.sta0004	00:0a:52:07:03:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	5.08	4.65	4.120879	7.075472
1.1.sta0005	00:0a:52:07:be:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	3.04	3.70	4.093567	3.324100
1.1.sta0006	00:0a:52:07:71:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	3.59	4.82	3.714286	8.333333
1.1.sta0007	00:0a:52:07:a5:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	3.42	3.80	3.517588	6.547619

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 8 clients running VI (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

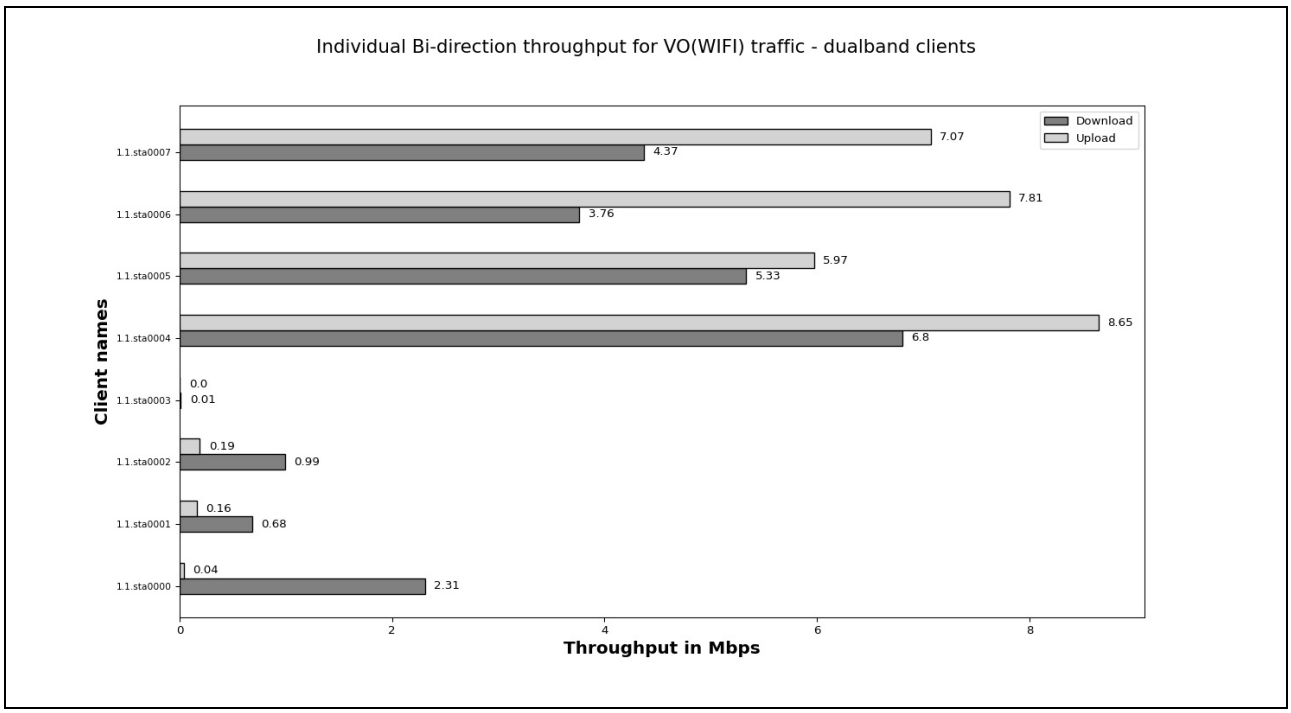


TOS : Video

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:6f:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.91	55.555556	6.557377
1.1.sta0001	04:f0:21:94:b7:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.16	0.66	46.153846	9.859155
1.1.sta0002	04:f0:21:94:93:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.01	0.00	33.333333	10.526316
1.1.sta0003	04:f0:21:94:68:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.08	0.41	27.272727	8.823529
1.1.sta0004	00:0a:52:07:03:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	6.14	4.32	1.641138	2.033898
1.1.sta0005	00:0a:52:07:be:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	5.99	6.48	1.117318	0.389610
1.1.sta0006	00:0a:52:07:71:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	6.29	3.86	1.583710	2.653400
1.1.sta0007	00:0a:52:07:a5:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	8.21	6.91	1.592357	0.000000

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 8 clients running VO (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:6f:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.04	2.31	11.111111	0.000000
1.1.sta0001	04:f0:21:94:b7:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.16	0.68	9.090909	5.921053
1.1.sta0002	04:f0:21:94:93:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.19	0.99	10.000000	5.434783
1.1.sta0003	04:f0:21:94:68:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.01	14.285714	2.409639
1.1.sta0004	00:0a:52:07:03:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	8.65	6.80	0.406918	0.632911
1.1.sta0005	00:0a:52:07:be:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	5.97	5.33	4.386839	1.232394
1.1.sta0006	00:0a:52:07:71:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	7.81	3.76	1.538462	1.773050
1.1.sta0007	00:0a:52:07:a5:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	7.07	4.37	2.054054	0.178891

Information	contact	support@candelatech.com